Inference Homework

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1) $SE = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{10^2}{100}} = \sqrt{\frac{100}{100}} = 1$

$$\left|\frac{\bar{X} - \mu}{SE}\right| = \left|\frac{52 - 50}{1}\right| = 2$$

$2 > 1.645$
$2 > 1.960$
$2 > 2.576$

The z-statistic of 2 is greater than 1.645 and 1.960 but not 2.576 so it is significant at the $\alpha = 0.1, 0.05$ levels but not $\alpha = 0.01$.

$\bar{X} - 1.960 \times SE \leq \mu \leq \bar{X} + 1.960 \times SE$

$52 - 1.960 \times 1 \leq 50 \leq 52 + 1.960 \times 0.1$

$50.04 \leq 50 \leq 52.196$

2)

90%:

$\hat{S} - 1.645 \times 1.3 \leq S \leq \hat{S} + 1.645 \times 1.3$

$3.1 - 2.1385 \leq S \leq 3.1 + 2.1385$

$0.9615 \leq S \leq 5.2385$

95%:

$\hat{S} - 1.960 \times 1.3 \leq S \leq \hat{S} + 1.960 \times 1.3$

$3.1 - 2.548 \leq S \leq 3.1 + 2.548$

$0.552 \leq S \leq 5.648$

99%:

$\hat{S} - 2.576 \times 1.3 \leq S \leq \hat{S} + 2.576 \times 1.3$

$3.1 - 3.3488 \leq S \leq 3.1 + 3.3488$

$-0.2488 \leq S \leq 6.4488$

We would fail to reject the null at the 99% confidence level as the confidence interval contains the null hypothesis of zero. We would reject the null hypothesis for the 90% and 95% confidence level because the confidence interval doesn't contain the null hypothesis of zero.

3)

$$SE = \sqrt{\frac{3^2}{36}} = \sqrt{\frac{9}{36}} = \sqrt{\frac{1}{4}} = 0.5$$

$$\bar{X} - 1.960 \times 0.5 \leq \mu \leq \bar{X} + 1.960 \times 0.5$$

$$11.5 - 0.98 \leq 12 \leq 11.5 + 0.98$$

$$10.52 \leq 12 \leq 12.48$$

$$\left| \frac{\bar{X} - \mu}{SE} \right| = \left| \frac{11.5 - \mu}{0.5} \right| = |-1| = 1 > 1.960$$

Based on both the 95% CI and the two-sided Z-score of 1, it is plausible that the population mean is 12.

4)

$$\bar{X}_A - \bar{X}_B = 25 - 22 = 3 \qquad \mu_A - \mu_B = 0$$

$$SE = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}} = \sqrt{\frac{64}{100} + \frac{49}{100}} = \sqrt{\frac{113}{100}} \approx 1.06$$

$$\left| \frac{(\bar{X}_A - \bar{X}_B) - (\mu_A - \mu_B)}{SE} \right| = \left| \frac{3 - 0}{1.06} \right| = \left| \frac{3}{1.06} \right| = 2.83$$

$$\bar{X}_A - \bar{X}_B - 1.960 \times SE \leq \mu_A - \mu_B \leq \bar{X}_A - \bar{X}_B + 1.960 \times SE$$

$$3 - 1.960 \times 1.06 \leq \mu_A - \mu_B \leq 3 + 1.960 \times 1.06$$

$$3 - 2.078 \leq \mu_A - \mu_B \leq 3 + 2.078$$

$$0.92 \leq \mu_A - \mu_B \leq 5.08$$

5)

$$SE = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{\frac{36}{81} + \frac{25}{64}} = \sqrt{0.444 + 0.391} = \sqrt{0.835} = 0.91$$

$$\left| \frac{\bar{X}_A - \bar{X}_B - 0}{SE} \right| = \left| \frac{2}{0.91} \right| \approx 2.198$$

6)

$$SE = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}} = \sqrt{\frac{2.25}{49} + \frac{1.96}{64}} = \sqrt{0.046 + 0.031} = 0.277$$

$$\left| \frac{\bar{x}_A - \bar{X}_B - 0}{SE} \right| = \left| \frac{3.2 - 2.8 - 0}{0.277} \right| = \left| \frac{0.4}{0.277} \right| \approx 1.44 > 1.960$$

$$\bar{X}_A - \bar{X}_B - 1.960 \times SE \leq \mu_A - \mu_B \leq \bar{X}_A - \bar{X}_B + 1.960 \times SE$$

$$0.4 - 1.960 \times 0.277 \leq 0 \leq 0.4 + 1.960 \times 0.277$$

$$0.4 - 0.543 \leq 0 \leq 0.4 + 0.543$$

$$-0.143 \leq 0 \leq 0.943$$

That our test stat of 1.44 is less than 1.960 and zero is within the confidence interval means that the difference in defect scores between the two factories is not statistically large relative to the noise at $\alpha = 0.05$.